Explore-then-Commit Algorithms for Decentralized Two-sided Matching Markets

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***Abstract*—THIS PAPER IS ELIGIBLE FOR THE STUDENT PAPER AWARD. Online learning in a decentralized two-sided matching markets, where the demand-side (players) compete to match with the supply-side (arms), has received substantial interest because it abstracts out the complex interactions in matching platforms (e.g. UpWork, TaskRabbit). However, past works [1]–[5] assume that the each arm knows their preference ranking over the players (one-sided learning), and the players aim to learn the preference over arms through successive interactions. Moreover, several (impractical) assumptions on the problem are usually made for theoretical tractability such as broadcast player-arm match ([1], [2], [5]) or serial dictatorship ([3], [4], [6]). In this paper, we study a decentralized two sided matching market, where we do not assume that the preference ranking over players are known to the arms apriori. Furthermore, we do not have any structural assumptions on the problem. We propose a multi-phase explore-then-commit type algorithm namely epoch-based CA-ETC (collision avoidance explore then commit) (`CA-ETC` in short) for this problem that does not require any communication across agents (players and arms) and hence decentralized. We show that for the initial epoch length of $T_\circ$ and subsequent epoch-lengths of $2^{l/\gamma}T_\circ$ (for the $l-$th epoch with $\gamma \in (0,1)$ as an input parameter to the algorithm), `CA-ETC` yields a player optimal expected regret of $\mathcal{O}[T_\circ(\frac{K\log T}{T_\circ\Delta^2})^{1/\gamma}\log(\frac{K\log T}{T_\circ\Delta^2}) + T_\circ(\frac{T}{T_\circ})^\gamma]$ for the i-th player, where T is the learning horizon, K is the number of arms and Δ is an appropriately defined problem gap. Furthermore, we propose a blackboard communication based baseline achieving logarithmic regret in T.** [1]

I. Introduction

Online matching markets (e.g. Mechanical Turk, Upwork, Uber, Labour markets, Restaurant) are economic platforms that connect demand side, (e.g. businesses in Mechanical Turk or Upwork, customers wanting Uber ride or restaurant reservations), to the supply side (e.g. freelancers in Upwork, or crowdworkers in Mechanical Turk, drivers in Uber, restaurant availability) ([7], [8]). In these platforms, the demand side (also known as the agent or player side) makes repeated decisions to obtain (match) the resources in the supply side (also known as arms side) according to their preference. The supply side is usually resource constrained, and hence it is possible that more than one players compete for a particular resource. Given multiple offers, the supply side arm chooses the agent of its choice. This agent is given the non-zero random reward, while all other players participated in the collision gets a deterministic zero reward. Usually the interaction between the supply and demand side is modelled as a bipartite graph, with players and resources in both sides have a preference (or ranking) over the other side, which is unknown apriori. Each agents' task is to learn this preference through successive but minimal interaction between the sides and thereafter obtain an optimal stable matching between the demand and the supply side.

[1]Appendix at https://bit.ly/ISIT_matchingmarkets

Multi-Armed Bandits (MAB) is a popular framework that balances exploration and exploitation and while navigating amongst uncertainty in the system [9], [10]. Learning in matching markets has received considerable interest in the recent past, especially from the lens of a multi-agent MAB framework [2]–[5]. In this formulation, the demand side corresponds to multiple players and the supply side resources correspond to multiple arms. The additional complexity here is the presence of competition among players. The objective of this problem is to learn the preference ranking *for both the players and arms simultaneously* through successive interactions. Once the preferences are learnt, matching algorithm like Gale-Shapley (Deferred Acceptance [11]) may be used to obtain the optimal stable matching.

Although the MAB framework in markets and bandits have been popular in the recent past, starting with a centralized matching markets (i.e., there exists a centralized authority who helps in the matching process, [1], [12]) to a more recent decentralized framework [3]–[5], [9], [13], *all* the previous works have several strong (and often impractical) assumptions on the market for theoretical tractability. We now discuss them in detail.

One-sided learning: One of the main assumptions made in all the previous works [1]–[4], [6], [13], including the state-of-the-art [5] is that of *one-sided learning*. It is assumed that the arms know their preferences over players apriori before the start of the learning, and hence the crux of the problem is to learn the preferences for the players only. So, the problem of two sided learning reduces to one sided learning. Here, algorithms based on Upper-Confidence-Bound [2], [3] as well as Explore-Then-Commit [4], [5] are both analyzed and sublinear regret guarantees are obtained. With the knowledge of the arms' ranking apriori, the system can correctly resolve the conflicts every time a collision occurs, which is used crucially in the analysis of such algorithms.

Additional structural market assumptions: Apart from the one-sided learning assumption, several additional structural assumptions on markets are usually being made to obtain sub-linear regret. In [2], [5], [6], it is assumed that when a player successfully receives a reward (or in other words, a player and an arm are successfully matched), the matched pair becomes a common knowledge to all the players, we refer this as (player, arm) broadcast. This assumption allows agents to communicate their learning progress. Moreover, [3], [6] assumed a *serial dictatorship* model where the preference ranking of players are assumed to be same for all the arms. In [2], the same assumption is termed as *global ranking*. In [13], this assumption is weakened and termed as α-*reducible* condition. Moreover, in [4], the authors propose a *uniqueness consistency* assumption on the market, implying that the leaving

participants do not alter the original stable matching of the problem.

A. Summary of Contributions

We study two learning algorithms, (i) Explore-then-Gale-Shapley (ETGS) in the presence of a blackboard, a communication medium, which we define shortly and (ii) epoch-based collision avoidance explore-then-commit `CA-ETC` a decentralized algorithm, which work for any *two-sided* market with *no restrictive assumptions*. By decentralized algorithm we mean that each agent performs actions based only on their past interaction in the market without any communication with other agents. Following we briefly explain our contributions

a) Decentralized two-sided learning algorithm: We take a step towards obtaining a provable two-sided learning algorithm for decentralized matching markets. To be concrete, we do not assume that the preference of the players are known to the arms apriori. Our proposed schemes, ETGS (with blackboard) and `CA-ETC` are based on Explore Then Commit (ETC) algorithm and are also *collision free*, similar to [5]. However, the algorithms are able to learn the preferences for both the agent and the arm side simultaneously through obtained samples in the exploration phase. Further, `CA-ETC` is decentralized. Note that in real world applications, like two-sided labor markets (Upwork, Uber, Restaurant), crowd-sourcing platforms (Mechanical Turk), scheduling jobs to servers (AWS, Azure) [14], [15], the preference of players side are apriori unknown and a two-sided learning algorithm is necessary.

We would like to point out a few very recent works on two-sided learning. [12] proposes a centralized algorithm for a market framework that allows utility transfer (monetary transfers), with convergence to a weaker notion of stability. [16] introduces algorithms with (player, arm) broadcasting and does not provide any regret guarantees. [17] assumes the existence of arm-sample efficient strategy which allows them to reduce their problem to one-sided learning after finitely many round-robin exploration rounds.

b) No structural assumptions on markets: We emphasize that our collision-free algorithms do not require any additional assumption on the economic market, like serial dictatorship, global knowledge of matching player arm pair, unique consistency etc, which makes them more practical. In applications like labor markets (Upwork, Taskrabbit) [18], crowd-sourcing platforms (Mechanical Turk), scheduling jobs to servers in an online marketplace (AWS, Azure) [14], [15], question answering platform (Quora, Stack Overflow) [19] the structural assumptions mentioned above are naturally not satisfied. So, there is a gap between theory and practice, and our proposed algorithms, is a natural first step towards closing this gap.

II. Problem Setting

We now explain the problem formulation. Consider a market with N players and K arms with $N \leq K$. Denote $\mathcal{N} = \{p_1, p_2, ..., p_N\}$ as the set of players and $\mathcal{K} = \{a_1, a_2, ..., a_K\}$ as the set of arms. In general, not all players and arms will participate in the market, but we consider that the participating agents include all the players and arms. At time step t, each player p_i attempts to pull an arm $A_i(t) \in \mathcal{K}$. When multiple players pull the same arm, only one player will successfully pull the arm, based on arm's preferences which are also learned over time.

Two-sided reward: Since our goal is to learn the preferences of the players and arms simultaneously, we propose a two-sided reward model in the following way. If player p_i successfully pulls an arm $A_i(t) = a_j$ then p_i receives a stochastic reward $X_j^{(i)}(t) \sim$ SubGaussian$(\mu_j^{(i)})$ (subGaussian with mean $\mu_j^{(i)}$ and unity subGaussian parameter), and arm a_j receives a stochastic reward $Y_i^{(j)}(t) \sim$ SubGaussian$(\eta_i^{(j)})$. We remark that for two sided learning, reward information for both the player and the arm side are necessary, and hence we propose this two sided reward model. If $\mu_j^{(i)} > \mu_{j'}^{(i)}$, we say that player p_i truly prefers arm a_j over $a_{j'}$. Similarly, if $\eta_i^{(j)} > \eta_{i'}^{(j)}$, we say that arm a_j truly prefers player p_i over $p_{i'}$. We denote $A_j^{-1}(t) := \{p_i : A_i(t) = a_j\}$ as the set of players proposing arm a_j, $\bar{A}_i(t)$ as the successfully matched arm of player p_i, $\bar{A}_j^{-1}(t)$ as the successfully matched player of arm a_j i.e. $\bar{A}_j^{-1}(t) \in \text{argmax}_{p_i \in A_j^{-1}(t)} \eta_i^{(j)}$. Then $\bar{A}_i(t) = A_i(t)$ when p_i is successfully accepted by arm $A_i(t)$ else if rejected $\bar{A}_i(t) = \emptyset$. When two or more players propose an arm a_j then only the most preferred player $\bar{A}_j^{-1}(t)$ among $A_j^{-1}(t)$ gets an reward $X_j^{(\bar{A}_j^{-1}(t))}(t)$ and other get a zero reward. Denote $M(t) := \{(i, \bar{A}_i(t) : i \in [N]\}$ as the final matching at round t. We assume that all the participating agents have strict preference ranking i.e. $\mu_j^{(i)} \neq \mu_{j'}^{(i)}$ and $\eta_i^{(j)} \neq \eta_{i'}^{(j)}$ for all arms $a_j \neq a_{j'}$ and players $p_i \neq p_{i'}$.

a) Stable matching: We seek to find stable matching [11], which is shown to be closely related to the notion of Nash equilibrium in game theory. A market matching is stable if no pair of players and arms would prefer to be matched with each other over their respective matches. Formally, $M(t)$ is stable if there exists no player-arm pair (p_i, a_j) such that $\mu_j^{(i)} > \mu_{\bar{A}_i(t)}^{(i)}$ and $\eta_i^{(j)} > \eta_{\bar{A}_j^{-1}(t)}^{(j)}$, where we simply define $\mu_\emptyset^{(i)} = -\infty$ and $\eta_\emptyset^{(j)} = -\infty$ for each $i \in [N], j \in [K]$. Let $M := \{m : m \text{ is a stable matching}\}$ be the set of all stable matching and define player-optimal stable matching $\bar{m}_p = \{(i, \bar{m}_i) : i \in [N]\} \in M$ as the players' most preferred match i.e. $\mu_{\bar{m}_i}^{(i)} \geq \mu_{m_i}^{(i)}$ for any $m \in M$ and for all $i \in [N]$. One can similarly define arm-optimal stable matching $\bar{m}_a = \{(\bar{m}_j, j) : j \in [K]\} \in M$ as the arms' most preferred one match i.e. $\eta_{\bar{m}_j}^{(j)} \geq \eta_{m_j}^{(j)}$ for any $m \in M$ and for all $j \in [K]$. Previous works [5] has also defined the notion of player-pessimal stable matching defined as the players' least preferred match $\underline{m}_p = \{(i, \underline{m}_i) : i \in [N]\} \in M$ i.e. $\mu_{\underline{m}_i}^{(i)} \leq \mu_{m_i}^{(i)}$ for any $m \in M$ and for all $i \in [N]$. Similarly we define arm-pessimal stable matching $\underline{m}_a = \{(\underline{m}_j, j) : j \in [K]\} \in M$ as the arms' least preferred match i.e. $\eta_{\underline{m}_j}^{(j)} \leq \eta_{m_j}^{(j)}$ for any $m \in M$ and for all $j \in [K]$.

b) Regret: Based on the different notions of stable matching, we define player-optimal regret for each player p_i, $i \in [N]$ over T rounds as

$$\overline{RP}_i(t) = \sum_{t=1}^{T} \mu_{\bar{m}_i}^{(i)} - \mathbb{E}[\sum_{t=1}^{T} X_j^{(i)}(t)]$$

and arm-pessimal regret for each arm a_j, $j \in [K]$ as follows

$$\underline{RA}_j(t) = \sum_{t=1}^{T} \eta_{\underline{m}_j}^{(j)} - \mathbb{E}[\sum_{t=1}^{T} Y_i^{(j)}(t)].$$

One can similarly define player-pessimal and arm-optimal regret. Note that the player-pessimal (arm-pessimal) regret is upper

	Two-sided	Assumption	Regret Type	Regret
[1]	No	centralized	player-pessimal	$\mathcal{O}(NK^3\log T/\Delta^2)$
[2]	No	(player, arm) broadcast	player-pessimal	$\mathcal{O}\Big(\frac{N^5K^2\log^2 T}{\epsilon^{N^4}\Delta^2}\Big)$
[3]	No	serial dictatorship	player-optimal	$\mathcal{O}\big(NK\log T/\Delta^2\big)$
[4]	No	uniqueness consistency	player-optimal	$\mathcal{O}\big(NK\log T/\Delta^2\big)$
[13]	No	α-reducible	player-optimal	$\mathcal{O}\big(\mathcal{C}NK\log T/\Delta^2\big)$
[5]	No	(player, arm) broadcast	player-optimal	$\mathcal{O}\big(K\log T/\Delta^2\big)$
[16]	Yes	(player, arm) broadcast	–	No Guarantee
This paper (Blackboard)	Yes	blackboard (eqv. to (play, arm) broadcast)	player-optimal	$\mathcal{O}\big(K\log T/\Delta^2\big)$
This paper (CA-ETC)	Yes	no assumptions	player-optimal	$\mathcal{O}\Big(T_\circ\Big(\frac{K\log T}{T_\circ\Delta^2}\Big)^{1/\gamma}+T_\circ\Big(\frac{T}{T_\circ}\Big)^{\gamma}\Big)$

TABLE I

TABLE COMPARING THE REGRET BOUND OF CA−ETC WITH EXISTING WORKS. HERE, N IS THE NUMBER OF PLAYERS, K IS THE NUMBER OF ARMS, T IS THE LEARNING HORIZON, Δ IS THE MINIMUM GAP (TO BE DEFINED LATER). ALSO, ϵ (IN [2]) AND $\mathcal{C}$ (IN [13] ARE PROBLEM DEPENDENT HYPER-PARAMETERS. FOR OUR ALGORITHM, CA−ETC, $T_\circ$ IS THE INITIAL EPOCH LENGTH AND $\gamma\in(0,1)$ IS AN INPUT PARAMETER RELATED TO THE SUBSEQUENT EPOCH LENGTHS.

bounded by player-optimal (arm-optimal) regret, and hence any upper bound on player-optimal regret automatically serves as an upper bound on player pessimal regret. When there are more than one stable matching i.e. $|M|>1$, the difference between player-pessimal and player-optimal regret can be $\mathcal{O}(T)$ due to a constant difference between $\mu^{(i)}_{\underline{m}_i}$ and $\mu^{(i)}_{\bar{m}_i}$, similarly for arms. Throughout we give guarantees for player-optimal regret, but the same hold for arm-optimal regret. We will henceforth refer to agents as players and arms i.e. all the participating agents in the market.

III. ALGORITHMS FOR TWO-SIDED MATCHING MARKETS

Gap: We define the gap of player p_i as $\Delta^{(i)}=\min_{j\neq j'}|\mu^{(i)}_{j'}-\mu^{(i)}_{j}|$ and gap of arm a_j as $\Delta'^{(j)}=\min_{i\neq i'}|\eta^{(j)}_{i'}-\eta^{(j)}_{i}|$. The universal gap is defined as the minimum of all the player and arm gap i.e. $\Delta=\min_{i\in[N],j\in[K]}\{\Delta^{(i)},\Delta'^{(j)}\}$.

We present the learning algorithms in this section. Before this we first note that, when the gap Δ is made common knowledge to every agent, then a logarithmic regret can be obtained, as the gap knowledge suffices to estimate the high-probability upper bound on the sufficient number of exploration round. Such direct estimation without any communication is not possible. Thus we start with a simple setup, where the participating agents are allowed to communicate using a blackboard then we present our main algorithm, CA−ETC.

A. Warmup: Learning with Blackboard

The blackboard model of communication is standard in centralized multi-agent systems, with applications in distributed optimization, game theory, and auctions [20]–[22]. We show that this is equivalent to broadcasting in the centralized framework (more on this in Appendix).

Definition 1: A blackboard $\mathcal{B}$ is a Boolean array of size $N+K$ where for $i\in\{1,...,N\}$, i^{th} entry is changed only by player p_i and for $j\in\{1,...,K\}$, $(N+j)^{\text{th}}$ entry is changed only by arm a_j. The blackboard can be viewed by all the players and arms.

The Algorithm 2 is Explore-then-Commit type where each agent performs exploration in a round-robin manner (thus avoiding collision) till everyone finds correct preference ranking which is communicated to other agents by blackboard and then commits to the stable-matching Gale-Shapley algorithm. We now explain the stages in detail.

∘ *Exploration Phase*: We first emphasize that when arms are unaware of the preferences over players, they are unable to resolve conflicts between colliding players. This necessitates a round-robin exploration, where at every round, each player proposes to a distinct arm and hence avoids the collision. To perform round-robin exploration, an index estimation scheme is proposed which assigns each player in a *decentralized* manner a distinct index.

Algorithm 1: Index Estimation (view of player p_i)

Input : arbitrary preference ranking of players for arm a_1
for round $t=1,2,\ldots,N$ **do**
  $A_i(t)=a_1$
  if $\bar{A}_i(t)=A_i(t)=a_1$ **then**
    Index = t;
  end
end

Algorithm 2: ETGS (player p_i) (Blackboard)

Perform Index estimation (Algorithm 1), $t=1$
while $\sum_{l=1}^{N+K}\mathcal{B}[l]\neq N+K$ **do**
  $A_i(t)=a_{(\text{Index}+t-1)\%K+1}$ // exploration
  Observe $X^{(i)}_{A_i(t)}(t)$ and update $\hat{\mu}^{(i)}_{A_i(t)},T^{(i)}_{A_i(t)}$ if $\bar{A}_i(t)=A_i(t)$
  Compute $\texttt{UCB}^{(i)}_k(t)$ and $\texttt{LCB}^{(i)}_k(t)$ for each $k\in[K]$
  if $\exists\sigma$ such that $\texttt{LCB}^{(i)}_{\sigma_k}(t)>\texttt{UCB}^{(i)}_{\sigma_{k+1}}(t)$ for any $k\in[K-1]$ **then**
    Preferences = σ and $\mathcal{B}[j]=1$
  end
  $t\leftarrow t+1$
end
// Gale-Shapley with $\sigma=(\sigma_1,...,\sigma_K)$
Propose using σ till acceptance
Initialize $s=1$
while $t\neq T$ **do**
  $A_i(t)=a_{\sigma_s}$
  $s=s+1$ if $\bar{A}_i(t)==\emptyset$
  $t\leftarrow t+1$
end

◇ *Index estimation*: In the proposed Algorithm 1, each player proposes to arm a_1 till it gets accepted and assigns itself the index as the round of acceptance. Thus, this phase lasts for N number of rounds. Intuitively, the index of each player will be the player's preference rank for arm a_1. In the beginning arm a_1 will have arbitrary but distinct preference ranking over players, hence leading to a distinct index for each player.

Based on the index, each player now performs round-robin exploration and each arm accepts the proposal. After every exploration round, each player p_i computes $\texttt{UCB}^{(i)}_k$ and $\texttt{LCB}^{(i)}_k$

for every arm $a_k, k \in [K]$ and similarly each arm computes the confidence bounds for all players. From the view of player p_i, after observing the reward from the matched arm $A_i(t)$, it updates an estimate of mean reward $\hat{\mu}^{(i)}_{A_i(t)}$ and the observed time $T^{(i)}_{A_i(t)}$ which is defined as the number of times player p_i is matched with arm $A_i(t)$ initialized as $T^{(i)}_{A_i(0)} = 0$. At time t this is updated using

$$\hat{\mu}^{(i)}_{A_i(t)} = \frac{\hat{\mu}^{(i)}_{A_i(t)} T^{(i)}_{A_i(t)} + X^{(i)}_{A_i(t)}}{T^{(i)}_{A_i(t)} + 1}, \quad T^{(i)}_{A_i(t)} = T^{(i)}_{A_i(t)} + 1$$

Now every player p_i updates the UCB and LCB estimates of $\mu^{(i)}_k$ for each arm a_k as $\mathtt{UCB}^{(i)}_k(t) = \mu^{(i)}_k + \sqrt{\frac{2\log t}{T^{(i)}_k}}$ and $\mathtt{LCB}^{(i)}_k(t) = \mu^{(i)}_k - \sqrt{\frac{2\log t}{T^{(i)}_k}}$, where $\mathtt{UCB}^{(i)}_k(t)$ and $\mathtt{LCB}^{(i)}_k(t)$ are initialized as ∞ and $-\infty$ respectively, similarly each arm a_j updates the confidence bounds for each player.

◇ *Preference ranking check*: After computing the confidence bounds, each player and arm performs a check whether they have found the true preference ranking over the other side. From the view of player p_i, it requires iterating over all possible permutations of arm preferences until a preference ranking is found which has disjoint confidence intervals i.e. a ranking σ such that $\mathtt{LCB}^{(i)}_{\sigma_k} > \mathtt{UCB}^{(i)}_{\sigma_{k+1}}$ for any $k \in [K-1]$ which implies player p_i truly prefers a_{σ_1} over a_{σ_2} over a_{σ_3} and so on.

◇ *Communication using Blackboard*: If a player or arm finds such disjoint confidence intervals, they set their corresponding bit on the blackboard to 1. Agents then enter the exploitation phase when all the bits are set to 1. Note that this happens synchronously, i.e. every player and arm enters the exploitation phase in the same round.

○ *Exploitation Phase*: Now when every agent learns the true preferences over the other side, they perform a decentralized Deferred Acceptance Algorithm [11] which is a polynomial time algorithm lasting at most K^2 rounds. It finds optimal match for the proposing side and hence a pessimal match for the other side. In our algorithm, players are the proposing side, thus in each round, every player proposes to an arm based on the learned preference ranking till acceptance i.e. if rejected in the last round it proposes to the arm with the next preference rank. Similarly, each arm, based on the learned preference ranking resolves the conflict and chooses the best player. Players (arms) can thus incur non-zero player-optimal (arm-pessimal) expected regret for at most K^2 rounds, after which a zero expected regret is guaranteed.

Regret Guarantee: We define $\Delta^{(i)}_{\max}$ as the maximum stable regret suffered by p_i in all rounds which is $\Delta^{(i)}_{\max} = \mu_{i,\bar{m}_i}$.

Theorem 1 (Regret of Algorithm 2): Suppose every agent plays Algorithm (2) for T iterations. Then, player p_i incurs the regret of

$$\overline{RP}_i(T) \leq \left(N + \frac{64K\log T}{\Delta^2} + K^2 + \frac{2NK\pi^2}{3} \right) \Delta^{(i)}_{\max}.$$

A similar upper bound holds for arm-pessimal regret.

Remark 1 (Different terms): First term in the regret is from index estimation phase, second from a high-probability number of exploration rounds, third from Gale Shapley algrorithm and fourth from the SubGaussian concentration bound.

Remark 2: The regret order-wise matches with the regret in the one-sided learning case [4], [5].

Algorithm 3: Epoch-based CA−ETC (view of player p_i)

Input : $T_\circ$, parameter $\gamma \in (0,1)$ with $b = 2^{1/\gamma}$
Run Algorithm 1 for Index Estimation
for $l = 1,2,...$ **do**
  Base Algorithm (exploration = $2^l T_\circ$, horizon = $b^l T_\circ$)
end

Remark 3: The blackboard setting can be changed by the assumption of global knowledge of matched (player, arm) pair broadcast [2], [5]. Using this one can detect in a decentralized manner whether all the agents have estimated correct preference ranking similar to the monitoring round of [5] for one sided learning. We comment more on this in the Appendix.

B. Epoch-based CA−ETC

We now present the main algorithm, namely CA−ETC which doesn't require a blackboard and is decentralized and communication-free. Here we describe the player's learning procedure in detail (the arm's learning is in the Appendix).

○ *Main Algorithm:* The proposed Algorithm 3, CA−ETC is epoch based, with each epoch consisting of increasing interaction rounds, in which each individual performs round-robin exploration and commits using learned preferences to the optimal matching Gale-Shapley algorithm. Here we choose exponentially increasing rounds, however, a polynomial algorithm leads to similar regret bounds. As before, to perform round-robin exploration the Index estimation subroutine Algorithm 1 is used by each player to get a distinct index.

◇ *Inside an Epoch:* In every epoch l, Algorithm 4 performs $2^l T_\circ$ exploration rounds and $2^{l/\gamma} T_\circ$ exploitation rounds. The parameter γ controls the exploration-exploitation trade-off in an epoch. The exploration phase is identical to the blackboard based algorithm with reduced computation. Here, each agent computes the confidence bounds and performs preference ranking check for disjoint confidence intervals *only once* after the defined number of exploration rounds for the epoch. If the check for disjoint interval succeeds each agent uses this preference ranking and if it fails the agent uses some arbitrary preference ranking and enter the exploitation phase which is identical to the blackboard based algorithm.

The idea of the algorithm is that when the total samples from exploration rounds exceeds $\lceil \log(\frac{16K\log T}{T_\circ \Delta^2} + 1) \rceil$ (see [5]), with high probability, each agent can estimate the correct preference. In CA−ETC, this happens after a finitely many epochs. Thus after these many epochs, each agent only incurs a non-zero regret in the exploration phase and Gale-Shapley rounds.

Theoretical Guarantees: Here, we present the regret bounds for CA−ETC for player p_i. Similar regret can be obtained for arm a_j (we defer this to the appendix).

Theorem 2: Suppose every agent runs CA−ETC with initial epoch length $T_\circ$ and input parameter $\gamma \in (0,1)$. Then, provided the initial epoch satisfies $T_\circ \geq \left(\frac{32K\log T}{\Delta^2 (T-N)^\gamma} \right)^{\frac{1}{1-\gamma}}$, the (player-optimal) regret for player p_i is given by

$$\overline{RP}_i(T) \leq N\Delta^{(i)}_{\max} + T_\circ \left(\frac{32K\log T}{T_\circ \Delta^2} \right)^{1/\gamma} \log\left(\frac{64K\log T}{T_\circ \Delta^2} \right) \Delta^{(i)}_{\max}$$
$$+ 2T_\circ \left(\frac{T}{T_\circ} \right)^\gamma \Delta^{(i)}_{\max} + K^2 \gamma \log\left(\frac{T}{T_\circ} \right) \Delta^{(i)}_{\max} + \frac{2NK\pi^2}{3} \Delta^{(i)}_{\max}.$$

Algorithm 4: Base Algorithm: ETGS (player p_i)

Input : Exploration rounds $2^l T_\circ$, Horizon $b^l T_\circ$

1 **for** $t = \sum_{l'=1}^{l-1} 2^{l'} T_\circ + 1, \ldots, \sum_{l'=1}^{l-1} 2^{l'} T_\circ + 2^l T_\circ$ **do**

2 $\quad A_i(t) = a_{(\text{Index}+t-1)\%K+1}$ // Exploration

3 $\quad$ Observe $X^{(i)}_{A_i(t)}(t)$ and update $\hat{\mu}^{(i)}_{A_i(t)}, T^{(i)}_{A_i(t)}$ if $\bar{A}_i(t) = A_i(t)$

4 **end**

5 Compute $\texttt{UCB}^{(i)}_k$ and $\texttt{LCB}^{(i)}_k$ for each $k \in [K]$

6 **if** $\exists \sigma$ such that $\texttt{LCB}^{(i)}_{\sigma_k} > \texttt{UCB}^{(i)}_{\sigma_{k+1}}$ for any $k \in [K-1]$ **then**

7 $\quad$ Preferences = σ

8 **else**

9 $\quad$ Preferences = arbitrary

10 **end**

// Gale-Shapley with $\sigma = (\sigma_1, \ldots, \sigma_K)$

11 Propose using σ till acceptance

12 Initialize $s = 1$

13 **for** $t = 1, 2, \ldots, b^l T_\circ - 2^l T_\circ$ **do**

14 $\quad A_i(t) = a_{\sigma_s}$

15 $\quad s = s + 1$ if $\bar{A}_i(t) == \emptyset$

16 **end**

Remark 4: CA-ETC takes γ and $T_\circ$ as input parameters to be chosen by the learner. Typical values of γ would be $\{1/3, 1/4, 1/5\}$, which would imply a polynomial dependence on $\log T$ and a weakly increasing function of T. In particular, for $\gamma = 1/4$ we have

$$\overline{RP}_i(T) \leq \mathcal{O}\left(\left(\frac{K \log T}{T_\circ \Delta^2}\right)^4 \log\left(\frac{K \log T}{T_\circ \Delta^2}\right) + T_\circ \left(\frac{T}{T_\circ}\right)^{1/4}\right)$$

In comparison to one-sided learning papers [2], [3], [5], the regret incurred by CA-ETC is high. This can be attributed to the cost of two-sided assumption-free learning. The dependence on γ comes from the multi-phase nature of CA-ETC. On top of a worse dependence on $\log T$ and Δ, CA-ETC also has a (weakly growing) polynomial dependence on T. We believe with algorithms that allow structured collision and some weak form of communication, these high regret term can be cut down.

Remark 5 (Type of CA-ETC): We emphasize that here we use exponential increment, however polynomial increment i.e. $l^2 T_\circ$ exploration and $l^b T_\circ$ total rounds can be used. We observe a similar regret upper bound, with the presence of 2^γ in place $\log(2^\gamma)$. Explicit upper bound can be found in the appendix.

Remark 6 (Different terms): First term in the regret comes from the index estimation subroutine. The second term results from the round-robin exploration before every agent estimates the true preference ranking. The third and fourth term comes from the round-robin exploration and Gale-Shapley after the rank estimation. The last term results from the SubGaussian concentration bound.

Remark 7 (Choice of $T_\circ$): We have a condition that needs $T_\circ$ to be not too small. However, note that, unless the gap Δ is too small, the condition is rather-mild and is trivial when T is very large. Of course, the optimal choice of $T_\circ$ depends on the gap, Δ and hence not known to the learner apriori.

IV. Simulations

We consider a market setup with 5 players and 5 arms. The mean is sampled uniformly between $(0,1)$ without replacement. We plot cumulative regret v/s horizon in Fig. 1 for a single run for 10^6 interaction rounds. For CA-ETC exp and poly we consider $2^l T_\circ$ ($b^l T_\circ$) and $l^2 T_\circ$ ($l^b T_\circ$) exploration (total rounds) respectively. we plot player-optimal and arm-pessimal regret for some players and arms. Arm-pessimal regret can be negative as arm can match with better players during exploration round, we observe this effect for arm 4. We keep $\gamma = 0.4$ and $T_\circ = 500$. Our results indicate that CA-ETC learns the *true* preferences after finitely many epochs which is $l_{\max} = 3$ for both CA-ETC exp and poly. We observe that the type of CA-ETC has an impact on the performance due to different sensitivity to parameter γ. In practice, we propose to use sufficiently larger $T_\circ$ for a good exploration warm-up and values of γ stated in Remark 4. We refer to additional experiments in the appendix.

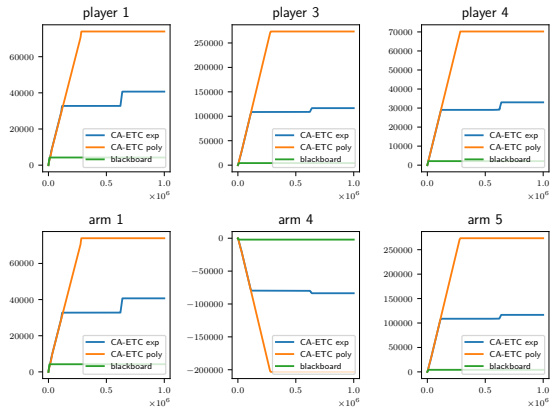

Fig. 1. 5x5 market cumulative regret plot

V. Discussion and Future Work

In this paper, we propose practical algorithms for two-sided learning in matching markets without restrictive assumptions, one with communication (blackboard) and other completely decentralized CA-ETC. Without any communication (implicit or explicit), and prior knowledge of Δ, logarithmic regret in two-sided learning seems unachievable. Prior work [3] uses implicit communication using structured collisions in one-sided learning, however, it is not beneficial when the arms cannot resolve conflicts correctly. This is due to the inherent asymmetry in the learning of two-sided matching, the non-proposing side (arm) need to learn the preferences before proposing side (players) for this communication to be useful. Our algorithm CA-ETC is symmetrical and synchronous i.e. both sides learn in similar fashion and enter commit phase at the same time. One future direction is to make the algorithm asynchronous, which can be done when a subset of the market preserves the stable match. Further, Algorithm 4 in Line 9 uses arbitrary preferences, which an agent based on prior knowledge and estimated confidence bounds can use a ranking close to *true* preference ranking and can thus lead to sharper regret bounds. In future, we would also like to study the market setup, with transferable utilities (i.e., monetary transfer) in a two-sided setting. Furthermore, markets are seldom static, and the preference ranking changes over time and capturing the dynamic behavior of markets in an assumption free two-sided setting is certainly challenging.

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